

Original research article

An adaptive RGB colour enhancement formulation for logarithmic image processing-based algorithms



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ARTICLE INFO

Article history:

Received 13 August 2016

Received in revised form 9 August 2017

Accepted 28 September 2017

Keywords:

Logarithmic image processing

Illumination normalization

Colour enhancement

Adaptive colour correction

Nonlinear colour space models

Generalized un-sharp masking

ABSTRACT

This paper presents an effective colour enhancement framework for statistical and logarithmic image processing (LIP)-based enhancement algorithms. The proposed approach utilizes the fusion of partial, multiple computed luminance channels with colour image channel statistics obtained from the input colour image for adaptive colour enhancement. The proposed scheme does not modify the image intensity channel, avoiding colour fading typically observed in colour images processed with conventional algorithms. The colour enhancement scheme compensates for the weaknesses of greyscale-based contrast enhancement and illumination normalization algorithms by focusing on preserving/restoring or enhancing colour. The proposed system avoids the conversion to complex, non-linear colour spaces such as HSI and HSV while producing similar results without manual adjustment of parameters. Additionally, an adaptive scheme for detection of images with unbalanced colour and uneven illumination is combined with the proposed system. Results show that the proposed scheme augments colour results of greyscale-based contrast enhancement algorithms and is relatively less complex compared to most algorithms in the literature.

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1. Introduction

The continued growth of high definition digital colour media and devices for their acquisition, transmission, processing and storage has precipitated the necessity for novel, effective and efficient colour image enhancement algorithms. These algorithms must utilize relatively low hardware resources to fit within the real-time digital hardware implementation constraints of embedded systems for such devices, which process great amounts of image and video data at high resolutions. Moreover, unwanted and unavoidable processes that manifest during the acquisition and transmission stages are still ever present in electronic devices. These include; blurring, noise, non-ideal illumination conditions, distortions, limitations of the dynamic range and display capabilities of electronic devices [1,2]. Several algorithms have been proposed to address these problems, however, the majority are suited to greyscale images.

Greyscale, histogram-based algorithms perform poorly when processing images with uneven illumination [1–4]. Conversely, standard logarithmic image processing approaches also yield subpar results and at best produce faded images when processing faded or bright, low contrast images [4,3]. In addition, these algorithms developed for greyscale image enhancement do not work well for RGB colour images due to non-linearity and vector nature of the RGB colour channels that comprise the image [1,2]. Thus, processing individual RGB image channels usually leads to colour distortion or fading and numerous schemes have been devised in the literature to rectify this deficiency. These include colour restoration functions in the

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Multi Scale Retinex with Colour Restoration (MSRCR) [5], Ratio Rule-based neural network guided Homomorphic filtering (HF) [6,7] and luma-based approaches utilizing non-linear colour spaces such as the HSI and HSV colour spaces [8,3,9]. The effectiveness of the non-linear colour spaces lie in their ability to decouple the Intensity or Value channel from the Hue and Saturation channels, which are all components embedded within the individual Red, Green and Blue image signals [1,2].

Currently, most recent work on colour image enhancement has focused on the non-linear colour space alternatives, which are consistent and proven in their performance. These include combining HSI combined with Homomorphic filters [8] and HSV colour spaces with Contrast Limited Adaptive Histogram Equalization (CLAHE) [10,11] and Genetic Algorithm or Particle Swarm Optimization (PSO) techniques [12], Dynamic Stochastic Resonance (DSR)-based image enhancement [13,14] and Tone mapping algorithms [15–19], Generalized Unsharp Masking (GUM) [11] and Probability-based illumination normalization [20] algorithms. However, algorithms such as Generalized Histogram Equalization (GHE) and Adaptive Histogram Equalization (AHE) [2] yield worse results in HSI or HSV space, leading to the gamut problem described by Naik and Murthy in their work [21]. They also proposed a technique for avoiding gamut problem after enhancement without resorting to colour coordinate transforms. The recent work on fast hue and range preserving histogram specification (HS) functions by Nikolova and Steidl [22,23], utilize these foundations for their proposed improved algorithm. Additionally, the proposed LIP algorithm by Patrascu and Buzuloiu [24], is also hue preserving, though its adaptive nature requires the solution of a set of relatively complex equations to obtain particular image parameters for enhancement. Alternative colour enhancement and preserving techniques include tensor-based methods using a Quaternion Discrete Fourier Transform [25], histogram modification equalization-based methods in 1D [26] and 3D spaces [27], development of a colour LIP colour correction framework [28] and multi-exposure homomorphic filtering [17]. Recently, works utilizing alternative LIP models [29] combining the Generalized Un-sharp Masking (GUM) [11,29] and probabilistic models for simultaneous illumination and reflectance estimation have been reported [20].

All the aforementioned alternatives still require colour conversion to HSV/HSI colour space to avoid/minimize the distortion or colour fading resulting from illumination correction operations in RGB space. This also adds increased hardware resource usage in digital hardware system implementations [3,30]. Furthermore, even when some of the algorithms are implemented in the HSI and HSV colour space, there is usually not much colour improvement in addition to faded colours observed after processing. Furthermore, any unregulated modification of the hue and saturation affects the final colour aesthetics of the colour image when converted back to RGB space. Thus, it is vital that the proposed system avoids the distortion of colours.

The primary motivation of the proposed algorithm in this work was to address the issues of colour distortion, automation and consistency of results in implementation. This is because most of the conventional approaches yield inconsistent colour results, are highly complex and not usually adaptive or automated, requiring manual adjustments of several parameters and selection of various functions. Even with canonical values, some of the earlier schemes do not yield best results for all images, lacking generality in application. In contrast, the proposed system yields consistent results as expected due to its theoretical foundations supported by extensive experimentation.

The rest of the paper is outlined as follows; the second section highlights the key contributions, features and objectives of the proposed scheme. The third section discusses relevant related works in detail and their limitations. The fourth section discusses the details of the proposed colour enhancement function and the justification for its implementation with presentation of theoretical formulation. The fifth section presents experimental results of the proposed scheme and its colour augmentation ability in addition to comparisons with other algorithms from the literature. Additional presentations include the modification of a previously computational complex algorithm by replacing a non-linear colour space converter with the proposed colour enhancement algorithm. Additional replacement of the edge preserving component of the contrast enhancement algorithm with a less complex but more effective alternative, reduced the run-time and complexity of the algorithm. The sixth section presents a brief discussion and possible applications and hardware realization possibilities of the proposed system. The final section presents the conclusions and a comprehensive summary of the work.

2. Key contributions, features and objectives

The key and novel contributions of this work include:

- Adaptive, combined fusion of intensity and value components to enhance colour rendition of processed colour images using weighted combination computed from colour channel statistics.
- Colour restoration of processed colour images with distorted or degraded colours using computed modal ratios from all three colour channels.
- Use of probability mass function (PMF) to compute standard deviation from mode (SDFM) and mean of modal ratio (MOM) parameters as inputs in the implemented system, ensuring adaptive and automated colour enhancement/restoration.
- Additional scheme for detecting dark images or images with dominant colour cast based on statistical contrast measure and entropy values used as inputs to a classifier for decision-based colour enhancement processing.

The key objectives and features include:

- The design and realization of a generalized and practical colour enhancement algorithm that can be a companion function to almost any grayscale-based image enhancement algorithm, improving its colour enhancement performance.
- Improving colour enhancement and colourfulness by regulated dilation without upsetting colour balance.
- Algorithm yields improved colour images, matching and/or surpassing results of HSI and HSV-based approaches while avoiding their complex and extensive colour coordinate forward and reverse transformation functions.
- The developed algorithm is fast, feasible and practical for hardware implementation due to its point-wise nature of operation.
- The proposed algorithm can also be modified to mimic results of any colour space conversion technique without performing the actual full colour space conversion.

3. Prior related works

The illumination correction algorithms treated in this paper are limited to the LIP and SLIP models with some discussions about the log-ratio (LR) based models. However as noted, these algorithms generally have a number of limitations, which include; minimal edge and contrast enhancement, inability to perform consistent colour correction, loss of details in bright image regions and inability to enhance faded, low contrast images. The colour fading and distortion aspect is addressed in this work and solutions to the colour fading problem from the literature are included for comparison. Furthermore, the drawbacks of these approaches are also briefly discussed with added references for further details.

3.1. Logarithmic image processing (LIP) and symmetric LIP (SLIP) models

The illumination correction term discussed here is based on a variant of the log-ratio (LR), log image processing (LIP) [11] and symmetric LIP (SLIP) model [29]. The Retinex-based algorithms are derived from the LR method while the Homomorphic filter is a variation of the model. The LR and the LIP values are defined in the range: $(0, L-1)$, where L is the number of defined grey levels. The operations for the LIP model [11,29] are;

$$\psi(a) = -R \ln \left(1 - \frac{a}{R} \right); a \oplus b = a + b - \frac{ab}{R}; \lambda \otimes a = R - R \left(1 - \frac{I_i(x, y)}{R} \right)^\lambda \quad (1)$$

$$\ominus b = -1 \otimes b = -R \left(\frac{b}{R-b} \right); a \ominus b = a \oplus (\ominus b) = R \left(\frac{a-b}{R-b} \right) \quad (2)$$

In (1) and (2), $\oplus$, $\ominus$ and $\otimes$ are vector addition, vector subtraction and scalar multiplication operations in the LIP domain, respectively. The terms a and b are vectors while λ is a scalar value. R is the range of the image vector, defined in the domain, $(0, R)$ and $\psi(a)$ is a LIP-based transformation function for vector a . Conversely, the operations for the SLIP model [29] are;

$$\psi(a) = -R * \text{sgn}(a) \ln \left(1 - \frac{|a|}{R} \right); a \boxplus b = R * \text{sgn}(a+b) \left[1 - \left(1 - \frac{|a|}{R} \right)^{\gamma_1} \left(1 - \frac{|b|}{R} \right)^{\gamma_2} \right] \quad (3)$$

Where in (3), the exponents, $\gamma_1 = \frac{\text{sgn}(a)}{\text{sgn}(a+b)}$ and $\gamma_2 = \frac{\text{sgn}(b)}{\text{sgn}(a+b)}$. Also;

$$\lambda \boxtimes a = R * \text{sgn}(\lambda a) \left[1 - \left(1 - \frac{|a|}{R} \right)^{|\lambda|} \right]; \boxminus b = -1 \boxtimes b = R * \text{sgn}(-b) \left[1 - \left(1 - \frac{|b|}{R} \right)^{|-1|} \right] \quad (4)$$

$$a \boxplus b = a \boxplus (\boxminus b) = R * \text{sgn}(a + (\boxminus b)) \left[1 - \left(1 - \frac{|a|}{R} \right)^{\gamma_1} \left(1 - \frac{|(\boxminus b)|}{R} \right)^{\gamma_2} \right] \quad (5)$$

$$\psi^{-1}(a) = R * \text{sgn}(a) \left(1 - e^{-\frac{|a|}{R}} \right) \quad (6)$$

In (3)–(6), $\boxplus$, $\boxminus$ and $\boxtimes$ are vector addition, vector subtraction and scalar multiplication operations in the SLIP domain [29], respectively. The signum function, sgn is defined as; $\text{sgn}(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$. The terms a and b are vectors while λ is a scalar value. R is the range of the vector, defined in the domain, $(0, R)$ and $\psi(a)$ is a SLIP-based forward transformation function for vector a while $\psi^{-1}(a)$ is a SLIP-based inverse transformation function for vector a .

3.2. Generalized Un-sharp Masking (GUM) algorithm

The GUM algorithm proposed by Deng [11] is composed of three essential components namely an Edge Preserving Filter (EPF), Contrast Enhancement (CE) function and the Adaptive Gain Control function (AGC). The algorithm utilized for the EPF is the Adaptive Median Filter (AMF) [31] and the selected CEF was the Contrast Limited Adaptive Histogram Equalization (CLAHE) [10] algorithm. The LIP- and SLIP-GUM algorithms are defined by the expressions; $v_{LIP} = z \oplus [\gamma \otimes (x \ominus y)]$ and $v_{SLIP} = z \boxplus [\gamma \boxtimes (x \boxminus y)]$ where x , y , z and γ are the input image, EPF output, CEF output and AGC output respectively.

Based on the work from the current literature, the LIP-GUM [11] and the SLIP-GUM [29] algorithms show some promising results. However, this is to be expected due to the fact that these algorithms are a combination of several algorithms such as EPF, CE and AGC components. The outputs of these components are evaluated in the LIP and SLIP domains respectively. However, as pointed out in [20], the traditional log-based approaches lead to detail loss in the extracted reflectance component. Additionally, the traditional LIP GUM leads to out of range errors for some images and have to be corrected with the SLIP approach [29]. Furthermore, there is still loss of detail in the GUM outputs in addition to little colour improvement despite the fact that the operations are performed usually on the intensity component of the HSI or HSV transformed image. This is in addition to the AMF and CLAHE, which are highly computationally complex as well. Thus the exploration of comparable, less complex algorithms to replace the AMF in the GUM forms part of the improvements of the work. Consequently, we utilize relatively less complex but highly effective algorithms for the EPF component of the GUM.

3.3. Colour enhancement approaches

The key, relevant, related works on colour enhancement include the Colour restoration routine [5], ratio rule [6,7], non-linear colour spaces [2], discrete cosine transform (DCT)-based method [32] and hue preserving algorithms [21,23]. Each scheme has its own unique features resulting in advantages and disadvantages.

3.3.1. Colour restoration function (MSRCR)

The colour restoration function for the popular MSRCR [5] is given as;

$$C_i(x, y) = \beta \log_{10} \left[\frac{\alpha I_i(x, y)}{\sum_{i=1}^S I_i(x, y)} \right] \quad (7)$$

$$I_{MSRCR_i}(x, y) = G [C_i(x, y) \cdot I_{MSR_i}(x, y) + b] \quad (8)$$

$$I_{MSR_i}(x, y) = \sum_{s=1}^S w_s [\log_{10}(I_i(x, y)) - \log_{10}(I_i(x, y) * h_s(x, y))] \quad (9)$$

$$h_s(x, y) = Ke^{-\frac{x^2+y^2}{c_s^2}} \quad (10)$$

In Eqs. (7)–(10), $i = 1, \dots, N$ where N is the number of bands or channels in the input image, $s = 1, \dots, S$, where S is the number of scales, w_s is the weight of the scale, $I_i(x, y)$ is the i th band of the input RGB colour image, $h_s(x, y)$ is the Gaussian surround function or spatial filter kernel, $I_{MSR_i}(x, y)$ is the i th band of the output image of the MSR algorithm while $I_{MSRCR_i}(x, y)$ is the i th band of the final output image of the MSRCR algorithm after applying the colour restoration function. Lastly, α , β , G and b are constants which are termed canonical gain-offset parameters to obtain the best results for the MSRCR [5] which is highly computationally complex.

The colour restoration function in (7) utilizes decoupled, normalized R, G and B channels of the original image (the value being converted to the logarithm domain) to reduce effects of inter-channel dependency [5].

3.3.2. Ratio rule

The ratio rule was developed by Seow, et al. [7,6] and is a state-space associative memory-based neural network. The network employs a line-of-attraction-based representation to describe the RGB inter-channel colour relationships within the image [6,7]. After training the network with the initial colour arrangement prior to processing, the system recalls the initial colour structure of the processed image by iteration [6,7]. This information is used to recalibrate the colours of the processed faded RGB colour image, effectively performing restoration. The homomorphic filter [1,2] was the image enhancement algorithm used for the processing of the RGB colour image, leading to the faded result that required restoration. The Ratio Rule scheme is evidently computationally complex [33], and it is not clear what the implications would be if the size of the images increases considerably in addition to increase in iterative operations. Furthermore, the algorithm and its results were not extensively compared to existing colour enhancement alternatives. A similar colour retrieval and enhancement scheme is proposed by Iqbal, et al. for surveillance images [34].

3.3.3. Non-linear colour spaces

Luma-based approaches have become the norm in colour enhancement-based image processing, due to the established colour coordinate transforms and successes in colour segmentation applications utilizing non-linear colour spaces. Studies carried out using this solution have shown that the system is feasible and hardware architectures utilizing the schemes have been reported in the literature [8,9]. These include the HSI and HSV colour spaces and in the latter, the hue is defined as the rotation angle between the Red plane and the origin [1–3]. The saturation is the distance from any point in the colour space system to the colour surface [1–3], while the value (V) represents the intensity (or luminosity) channel attributes of the image. The equations for the RGB to HSV transformation [1,2] are given as;

$$V = \max(R, G, B); S = V - \min(R, G, B) \quad (11)$$

$$H = \begin{cases} \frac{G-B}{S}; & R = V \\ 2 + \frac{B-R}{S}; & G = V \\ 4 + \frac{R-G}{S}; & B = V \end{cases} \quad (12)$$

The HSI colour space was based on the human visual system and colour perception and is modelled with a colour hexagon, triangle or circle representation [1,2]. The hue, saturation and intensity attributes are similar in definition to that of the HSV system; however, their mathematical expressions are different. The primary and secondary colours of light are vertices in the hexagon model and line segments in the colour triangle model [1–3]. The primary and secondary colours are separated by a value of 120° in the colour circle representation [1–3]. The equations for the RGB to HSI conversion [1,2] are given as;

$$I = \frac{R+G+B}{3}; S = 1 - \frac{\min(R, G, B)}{I} \quad (13)$$

$$H = \begin{cases} \theta; & \text{if } B \leq G \\ 360^\circ - \theta & B > G \end{cases}; \theta = \cos^{-1} \left(\frac{(R-G) + (R-B)}{2\sqrt{((R-G)^2 + (R-B)(G-B))}} \right) \quad (14)$$

The HSI colour conversion yields superior results compared to the HSV colour space [30,3,8]. Large LUTs are normally used for the hue computation in HSI space, though this becomes impractical for images with larger bit depths greater than eight bits per pixel (bpp) on digital hardware systems with limited memory. There are numerous alternative equations for calculating the hue component [3,30], which is the most crucial aspect. The search for alternatives is partly due to the challenges encountered in accurately computing the hue channel in digital hardware devices such as FPGAs using fixed point [3,30]. However, a hardware architecture solution utilizing a proven alternative formula, resulting in the full HSI converter FPGA hardware architecture has been reported in recent literature [3,30]. To date, this is still the only publicly known or available implementation.

Based on work in the literature, the HSV variant is the most common digital hardware implementation of nonlinear colour space converters [9,3,35,18]. This is due to its relative ease of implementation compared to the HSI colour space conversion in fixed-point digital hardware [30]. Since the Hue is the most important aspect of the human visual system, on which the HSI and HSV colour spaces are based, any distortions in computation would affect the final result, even without further processing [3,30]. In the HSV case, hue computation is much more straightforward, using multiplexer and basic rational operations.

However, utilizing the non-linear colour space converters result in a speed performance bottleneck in hardware if only the intensity channel is processed (though this is an advantage in terms of reduced computational load and complexity, if the image processing algorithm itself is relatively highly involved). This is because latency increases since the Hue and Saturation channels must be synchronized with appropriate delays to ensure that they are correctly combined with the corresponding processed intensity channel for accurate conversion to RGB space after processing. This thwarts parallelization efforts due to the dependency of all three channels on each other for the reverse conversion. Even pipelining techniques reach a peak performance ceiling beyond which there is no further improvement in speed of the design. Latency is increased further along with throughput as a result but this delay adds to the speed of the reverse conversion. Thus, no matter how fast the image processing algorithm or the colour space conversions are, they all depend on the amount of delay required to ensure accurate synchronization and conversion to RGB space.

3.3.4. DCT-based colour enhancement

This scheme was proposed by Mukherjee and Mitra for colour enhancement in DCT domain [32]. The approach converts the RGB image to the YCbCr or YUV colour space and computes the DCT of each component [32]. Three main processes are then performed on these transformed components, namely; local background illumination adjustment, local contrast preservation and colour preservation [32]. In spite of its apparent simplicity, there is DCT computation, block splitting and merging steps and several parameters to consider in addition to a blocking artefact removal (BAR) algorithm [32]. Also, the number of parameters to tune increases, depending on the function chosen for illumination adjustment. Thus, the algorithm is not automated, requires user specified parameters, a selected function and option of using BAR algorithm if blocky effects appear in the processed images.

3.3.5. Hue and range preserving colour enhancement

Initially, the work by Naik and Murphy proposed a colour image enhancement algorithm that avoided the gamut problem [21]. Recently, Nikolova and Steidl proposed a fast hue and range preserving histogram specification algorithm to mitigate the gamut problem that occurs after RGB colour image enhancement [23]. Rather than processing each individual channel, they convert the image to the HSI colour space and map the intensity channel to an ideally chosen histogram using strict ordering algorithm developed by [36]. However, the algorithm is not automated and performs best with user specified parameters and may yield inconsistent results.

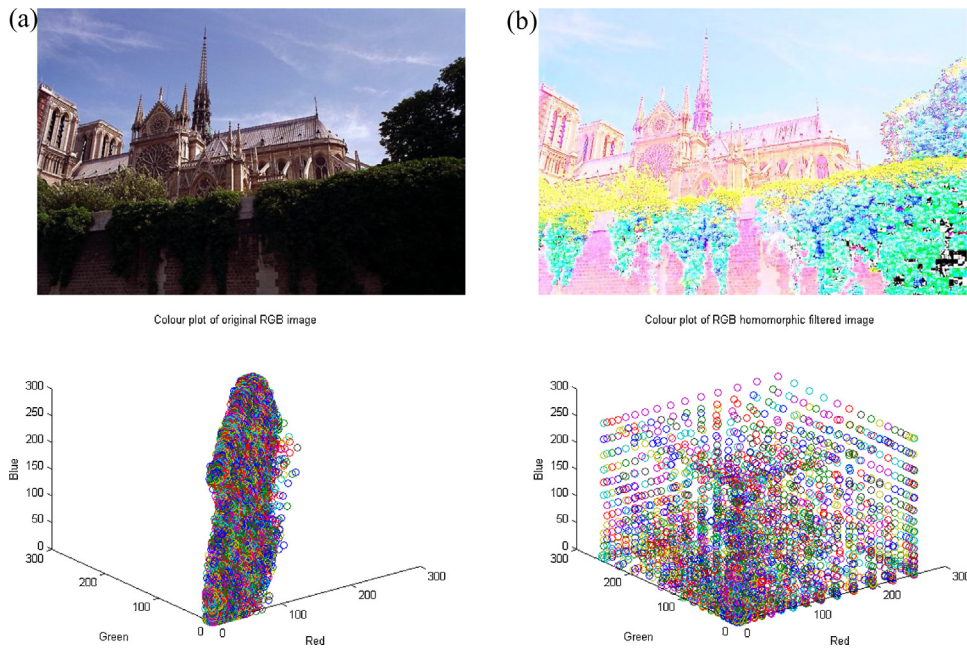


Fig. 1. (a) Original image and colour profile plot (balanced profile) (b) Colour distorted image and colour profile plot (unbalanced profile).

4. Proposed colour enhancement approach

Based on analysis of related works, there is need for a scheme that combines the effectiveness and advantages of the HSI and HSV colour space while avoiding their weaknesses. Such a scheme should be able to handle most images presented and yield good colour enhancement/rendition regardless of the damaging effects of the prior applied image contrast enhancement algorithm. This is in addition to avoidance of distortion or degradation of the image pixel intensity values in the process of colour rendition. The requirements of this system include;

- Perform on par or (if possible) better than the HSI and HSV colour converters
- Inherit their strengths (decoupling intensity, thus becoming contrast agnostic) and eliminating their weaknesses.
- Should not distort colour balance of the R, G and B channels.
- Suitable, practical and amenable to hardware implementation.
- Should be adaptive and avoid manual parameter adjustments.
- Should be generic and work with most algorithms it is combined with.

The general form of the proposed algorithm consists of a contrast enhancement transfer function, $T(f(x, y))$ and colour enhancement function, $C(f(x, y))$ of an input RGB colour image $f(x, y) = \{I_i(x, y); i = 1 \text{ to } N\}$ with the number of channels, $N=3$ expressed in the following form for each channel;

$$I_{i_{final}}(x, y) = C(I_i(x, y)) \cdot T(I_i(x, y)) \quad (15)$$

Assume the RGB colour cube as a container, within which all possible colour image configurations will fit. Thus, we show a colour plot of the relationship between the R, G and B channels in a colour image. If we consider the image pixels as particles, and that the cohesion of these particles taken as a function of their distances apart from their mean, then we can measure the correlation of the pixels to each other. Ideally, highly correlated particles are closer (greater cohesion) while those that are further apart have less correlation (lower cohesion) with each other. However, since the distribution of image colour pixels do not follow a normal distribution and will have highly skewed mean values, the usage of the mode in the probability mass function (PMF) gives a better estimate. Thus, we initially assume that the colourfulness or colour aesthetics that is visualized by the human visual system and interpreted by the brain could be perceived as a measure of the cohesion or correlation of these particles defined by their closeness to their modal ratio values. If the R, G and B particles are slightly apart, the colour balance is slightly modified. However, if the distance increases between the particles and the centre of the probability mass distribution, the colour fades or distorts and colour balance is lost (destabilization). This balance is the inherent colour relationship between the R, G and B channels of colour images. We proceed to test these assumptions with actual experiments to verify, modify, discard or re-evaluate them.

The figures shown in Fig. 1 illustrate this idea of a colourful (Notre Dame) [5] image and its distortion in relation to its RGB colour plot. In Fig. 1(b), the image is highly distorted since the R, G, B particles are scattered and some are almost at the

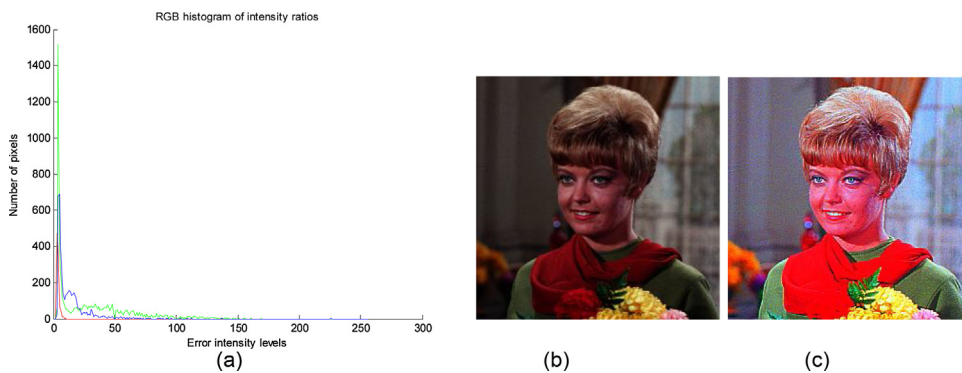


Fig. 2. (a) RGB histogram of intensity ratios of (b) original image (c) processed with adaptive modal ratio guided RGB spatial HF (modal R_ratio = 2.5, modal G_ratio = 3 modal B_ratio = 4).

Table 1

Sample RGB colour images and their modal ratios.

Image	Mode (R)	Mode (G)	Mode (B)	Mode (rounded) (R)	Mode (rounded) (G)	Mode (rounded) (B)
Girl [37] (a)	2.5	3	4	3	3	4
Swan [5] (b)	3	3	3	3	3	3
Notre Dame [5] (c)	∞	3	3	∞	3	3
StarTrek [5] (e)	4	2.5	3	4	3	3
RVs [38] (f)	3	3	3	3	3	3
Church [39] (h)	2	3	3	2	3	3
Sky lake [39] (i)	3.8	2.7143	2.7143	4	3	3
Parked car [39] (j)	3	3	3	3	3	3
Butterfly [39] (k)	2	3	4.5	2	3	5
Grandma & baby [39] (l)	3	3	3	3	3	3
Books1 [39] (m)	3.8	2.7143	3	4	3	3
Colours [39] (n)	2.75	3.1429	3.1429	3	3	3
Colour gamut [39] (o)	3	3	3	3	3	3
Scale models [39] (p)	3.8	2.7143	2.7143	4	3	3
Dawn.dusk [39] (q)	3	3	3	3	3	3
Mountain peak [39] (r)	3.0120	3	2.9880	3	3	3
Room [39] (s)	3	2.5714	4	3	3	4
Woman [39] (t)	3.5	3	2.6250	4	3	3
Cat [39] (u)	3.3333	2.8571	2.8571	3	3	3
Boy on stones [23] (v)	3	3	3	3	3	3
Cathedral2 [23] (w)	3	3	3	3	3	3
Orchid [23] (x)	3	3	3	3	3	3
Bungalow [23] (y)	3	3	3	3	3	3

edge of the cube. This distortion can arise from a grey-scale algorithm with extremely high gain incorrectly applied to a RGB colour image. Furthermore, the degradation is drastic in the high frequency regions or areas with more “energetic” pixels. Note that the smooth, slowly varying regions (blue sky) are relatively unaffected by the distortion.

The proposed system re-establishes correlation between the various colour channels by forcing these scattered particles closer or re-arranging the structure of the particles without upsetting the colour balance. This reduces the distance between particles, indicating a visual plot of colourfulness. The proposed method enhances colour images, avoids saturation of bright/dark regions of images, thus suitable for both contrast enhancement and illumination normalization/correction algorithms. We will show that considerable improvements are obtained with the proposed colour enhancement scheme.

4.1. Colour channel dominancy and distortion prediction

Initial experiments were performed to determine any discernible consistent pattern between the R, G and B channels using normalized decoupled versions of the channels based on formulations from the literature. The sample results for the initial experiment are as shown in Fig. 2(a) with the histogram of the inverse normalized RGB channels or intensity ratios of the girl image [37] processed with the intensity ratio controlled HF. Observing the modal ratios show that the Red channel has a less common occurring value. Any attempt at using the actual modal ratio values for the red component in this case will yield erratic results as shown in Fig. 2(b) as the most occurring pixel ratio is not up to a third of the RGB colour image. Thus, when using the adaptive the red channel is overcompensated, leading to oversaturation, causing the observed colour bleeding effect in Fig. 2(b). The red flower in the image is completely indistinguishable from the red scarf in Fig. 2(c) and other red regions are also overcompensated. Sample results in Table 1 show the modal ratios of the normalized colour channels

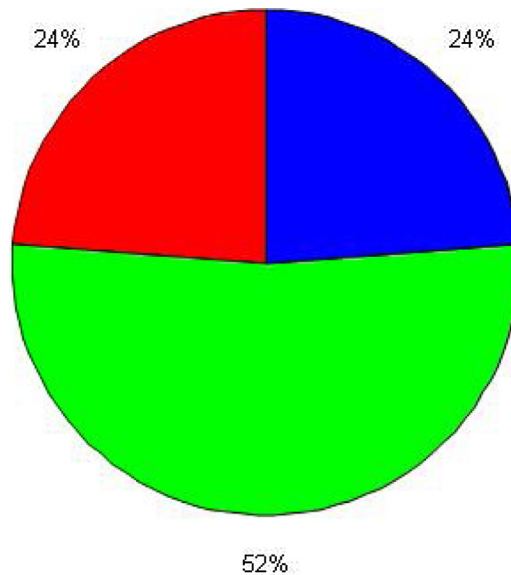


Fig. 3. Pie chart plots showing RGB modal ratio profiles of *Girl* image.

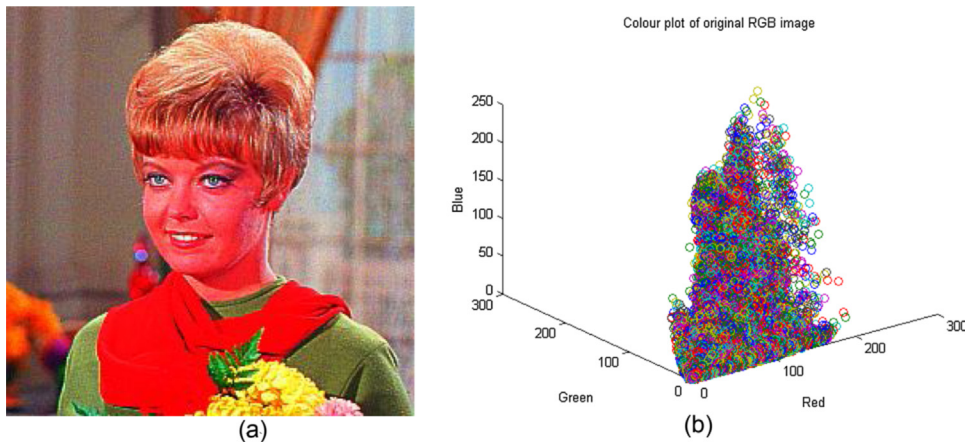


Fig. 4. (a) Image processed with HF and intensity-channel model (HSI) in RGB space (b) RGB colour plot.

of several tested images and indicate that most of the pixel values are close to the assumed model of intensity, with a factor of 3.

The pie charts generated in Fig. 3 shows the modal ratio profile (i.e. most common ratio) values obtained for the decoupled, normalized R, G and B channels of the *Girl* image. The values generated in the plots indicate the percentage of values that match the modal ratio of each channel. Rounding up of values as shown on the left hand side of Table 1 will increase the respective percentages for each of the channels in the modal profiles, which would lead to colour distortions after processing. Based on quick observation, 3 is the most commonly encountered ratio value, validating the naïve intensity model from the HSI by experimentation. The pie chart representation was chosen since it provides quick visual cues regarding the balancing of R, G and B channels of a colour image, which may not be easily observed using conventional visual representations such as histograms and image diagrams. The dominance of a channel can be easily deduced due to the pie chart plots of the modal ratio values in each channel (Fig. 4).

The reason for the distortion of colours in the background component of the image in Fig. 2(c) is due to the naïve adaptive scheme applying the computed modal ratio of each channel as a factor. This favours the less dominant channels (in this case, R and B) and will result in some colour regions of the image being over enhanced (where modal ratio values are less than or equal to 3) and punishes the more dominant channels (in this case, G) while other colour regions are slightly or moderately enhanced (where ratio values are greater than 3). This is confirmed by the saturated blue background and red hue of the face and scarf foreground and the RGB image modal ratio profile in Fig. 3. Thus, the modal ratio profile is a reliable indicator of the results of the blind, adaptive scheme, which are clearly undesirable.

This observation may possibly explain the gamut problem for certain images after processing in a different colour space and reverse-mapped to RGB.

4.2. RGB colour enhancement

Initially, the system was developed using the intensity component computation of the HSI colour space as a starting point since it provides the best colour rendition when combined with enhancement algorithms [30]. Additionally, the proposed system is less computationally complex than the colour restoration function and the ratio rule method (which is iterative). Based on approximations of the experimental results, we define the enhanced output colour image as;

$$I_{final}(x, y) = \left[\frac{I_i(x, y)}{I} \right] \cdot I_i(x, y); I = \sum_{i=1}^N w_i I_i \quad (16)$$

Where $i = 1 \dots N$; N being the number of channels ($N = 3$ for RGB image) and the intensity channel, I while w_i is the weighting factor of the i th channel such that $\sum_{i=1}^N w_i = 1$. The weights do not have to be equal but their sum must be unity. For the HSI colour space, the intensity is calculated and setting $R = I_1$, $G = I_2$, and $B = I_3$ and recognising that $w_1 = w_2 = w_3 = \frac{1}{3}$, the well-known form is as shown in (17)

$$I = \frac{I_1 + I_2 + I_3}{3} = \frac{R + G + B}{3} \quad (17)$$

Thus, based on extensive experimentation, a large majority of colour images are properly enhanced using the scheme, while a few outlier images need some modification by using unequal weights for the improved colour output. The well balanced images will yield ratios of 3 for each channel, but other images will yield ratios greater than or less than 3 for either the R, G or B channels. Investigation of the degree of dispersion of pixel ratios from the modal value necessitated a measure to evaluate the standard deviation from mode (SDFM) and is defined as;

$$SDFM = \rho = \sqrt{\frac{\sum_{i=0}^N (I_i - m)^2}{N}} \quad (18)$$

Where m is the modal value obtained from I_i , the image intensity ratio (or inverse normalized image) distribution of the i th channel of the colour image. The values obtained from the R, G and B channels of the girl image are 0.6711, 13.9831 and 6.7491 respectively. The figures imply that the R channel has the lowest variance from the modal value. However, this does not give a clear explanation, thus the usefulness of the modal profile, which indicates that the R channel intensity ratios are quite low. Evaluation of the maximum value in the R channel yields a value of 9 and a minimum value of 1.0233. For the G channel, maximum value is 168 and minimum value is 1.2857. For the B channel, maximum value is 225 and minimum value is 1.6667. Thus, the modal values actually give the maximum value of the probability mass function, which cannot be deduced from the maximum and minimum pixel intensity values across colour channels. The standard formula for computing a greyscale image from RGB space based on the luminous efficacy curve [1] is given as;

$$I = 0.299 * R + 0.587 * G + 0.114 * B \quad (19)$$

In (19), $w_1 = 0.299$, $w_2 = 0.587$, $w_3 = 0.114$ and in all cases $w_1 + w_2 + w_3 = 1$. This scheme leads to colour bleeding for images and high saturation in the overcompensated channel when using mainly the HSI intensity model. The HSI result has too much colour and using the colourfulness parameter by Susstrunk and Hasler [40], the output colourfulness is 174.6909 (which is quite high) and relative colour enhancement is 3.9549. It is clear from the image that the colourfulness measure alone cannot provide any information about colour aesthetics of balance as earlier assumed.

Due to this oversaturation of the red channel using the HSI colour space, the Value channel from the HSV model was also tested and is given as;

$$V = \max(R, G, B) \quad (20)$$

The results are shown in Fig. 5 with colourfulness value of 78.8977 and relative colour enhancement value of 1.7862. Though the HSV provides a better balance, there is still dominance of Red channel in regions of the image.

Thus a combination of values was devised to yield an image with more balanced colour output since the value channel would reduce the oversaturation effect of the intensity channel. We obtain the intensity-value (IV) result by combining the intensity and value components in the general form;

$$IV = \alpha I + \beta V \quad (21)$$

In (21), α and β are constants < 1 such that $\alpha + \beta = 1$. By setting $\alpha = \beta = 0.5$, we obtain the IV component as an equally weighted average of the two components as $IV = \frac{I+V}{2}$. However, more experiments may lead to cases where $\alpha < \beta$ for image with balanced colour output. The result of the combination with the HF is shown in Fig. 6. Comparing the image processed with HSI-HF and the RGB IV-HF shows the improvements of the model and its similarities with the HSI colour space model results in RGB space. Note that the red flower and scarf features are much more discernible in the IV-model than in the HSI results. Thus, combining both value and intensity schemes yields a hybrid model, which is better than either of the two, with more balanced saturation by combining the strengths of each colour space model without inheriting their weaknesses.

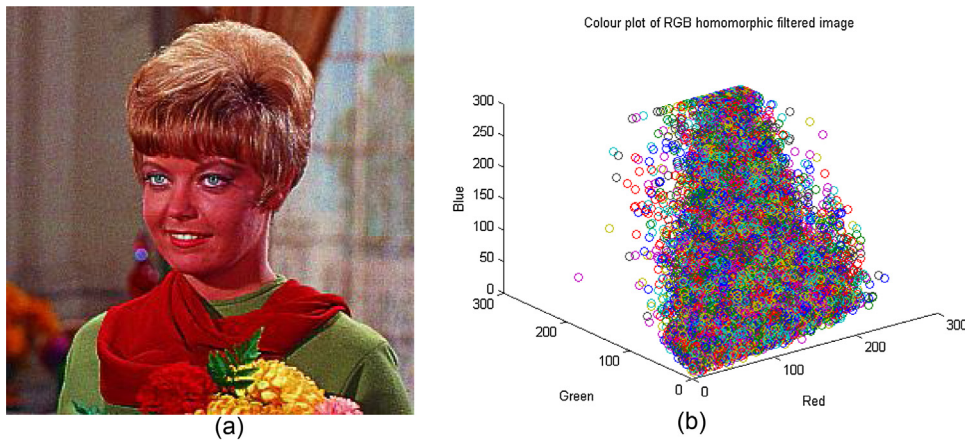


Fig. 5. (a) Image processed with HF and value-channel model (HSV) in RGB space (b) RGB colour plot.

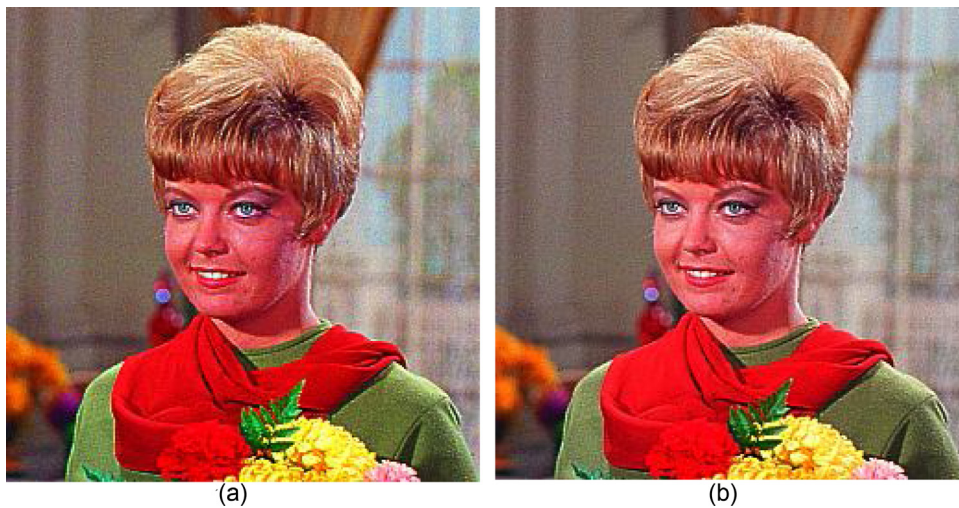


Fig. 6. Comparison of (a) combined intensity-value-channel version (RGB-IV-HF) and (b) HSI-HF.

It should be noted that the weighted average is one method of combining the channels and alternative methods include evaluating maximum of the intensity (I) and value (V) channels. However, by deduction and experiment, the latter results resembled the HSV results using maximum while a minimum yielded the HSI results. Other relationships such as multiplicative-square root relationship yielded good but inconsistent results. Thus, the weighted average model still provided the best and most consistent results. Based on experiments, the probability mass function (PMF) of intensity ratios obtains a maximum value around 3, which is the modal ratio value. With the results obtained, the complete colour enhancement routine can be redefined using the IV model in the general form, with the constants α and β as;

$$I_{i_{final}}(x, y) = \left[\frac{I_i(x, y)}{IV} \right] \cdot I'_i(x, y) = \left[\frac{I_i(x, y)}{\alpha I + \beta V} \right] \cdot I'_i(x, y) \quad (22)$$

Further expansion of the formula leads to the following left-hand-side (LHS) expression in (23), using the intensity and value channel calculations. If the RGB to intensity image conversion is to be employed with varying weights, the general form is shown in the right-hand-side (RHS) expression of (23);

$$I_{i_{final}}(x, y) = \left[\frac{I_i(x, y)}{\alpha \left(\frac{R+G+B}{3} \right) + \beta (\max(R, G, B))} \right] \cdot I'_i(x, y) = \left[\frac{I_i(x, y)}{\alpha \left(\sum_{i=1}^N w_i I_i(x, y) \right) + \beta (\max(R, G, B))} \right] \cdot I'_i(x, y) \quad (23)$$

4.3. Colour saturation moderation and control

As was earlier noted, the colours of some enhanced images are oversaturated. Consequently, we utilize the modal ratio to control the level of colour enhancement by utilizing the reciprocal as an exponent for the colour enhancement formula. In other words, control of the saturation is obtained by modifying Eq. (23) as;

$$I_{i_{final}}(x, y) = \left[\frac{I_i(x, y)}{\alpha \left(\sum_{i=1}^N w_i I_i(x, y) \right) + \beta (\max(R, G, B))} \right]^{1/m} \cdot I'_i(x, y) \quad (24)$$

For emulation of results from any other linear colour spaces (which are hue preserving as noted by Naim and Murthy), the expression becomes;

$$I_{i_{final}}(x, y) = \left[\frac{I_i(x, y)}{\left(\sum_{i=1}^N w_i I_i(x, y) \right)} \right]^{1/m} \cdot I'_i(x, y) \quad (25)$$

This would apply to the matrix-based, linear colour space transformations such as YIQ, YUV, YCbCr, etc, where the intensity/luminance notation can be appropriately substituted (though the expression may be of similar or greater complexity than the linear colour conversion operations). Finally, in practice, we select $m = 1$ or 2 for ease of operations and hardware implementation amenability, resulting in a linear or square root operation. The summary of the experiments and observations is as follows;

- The initial scheme utilizing only the weighted intensity model did not yield considerable colour balancing, which is achieved with saturation tuning in the HSI and HSV colour spaces.
- Certain colour images match the Intensity channel model of the HSI colour conversion and yield consistent results. The non-ideal colour images are more suited to the hybrid I–V model and the latter model yields good results for most colour images processed.
- The system avoids colour space transformation and enhances the colour improvement attribute of the algorithm it is combined with.
- The system targets the inherent relationship between the R, G and B channels and thus its generality is satisfied based on experiments, indicating that the relationship between R, G and B channels is a reasonably predictable one.
- Finally, it is evident that this scheme appears to take a form similar to that of the colour restoration function employed in the MSRCR and also similar to the Ratio Rule method, but with reduced complexity and no iterations or canonical parameters.

With the foundations of the colour enhancement operator elucidated, we proceed with experiments to fine tune the algorithm for outlier images. After that, we then proceed with the augmentation of several conventional image contrast enhancement and illumination correction algorithms, which usually perform poorly for colour images.

4.4. Dynamic computation of the m parameter for fully adaptive colour enhancement

It was discovered in the course of further experiments that setting the same value for the m parameter for each of the R, G and B channels results in inconsistencies for some images. This is due to the fact that the results of the m value are dependent on the nature of the algorithm and the image being processed. For example, for some images, $m = 1$ is adequate while for others it leads to over- or under-enhancement of colours. Conversely, for other images, a value of $m = 1$ or increasing the value of m resulted in faded colours. Thus, it was imperative to obtain a standard formulation that enabled the consistent enhancement of colours that were derived from the image attributes rather than a fixed parameter. Subsequently, other image statistical parameters were explored in order to obtain values for enhancement of colour images. The mean and standard deviation proved unreliable and yielded highly distorted results while the mode once more proved to be the measure of choice for best results. Initial attempts using the mode utilized the following formulation;

$$r = \frac{R + G + B}{R}; g = \frac{R + G + B}{G}; b = \frac{R + G + B}{B} \quad (26)$$

$$m_r = \text{mode}(r); m_g = \text{mode}(g); m_b = \text{mode}(b) \quad (27)$$

However, this formulation yielded inconsistent results for images with unequal modal ratios. As a result, some image results depicted colour enhancement or preservation/retention while others yielded faded colours, which was the drawback

of the earlier mentioned image enhancement algorithms. The next alternative was to utilize the result from (21) in the expression;

$$r = \frac{R}{IV}; g = \frac{G}{IV}; b = \frac{B}{IV}; IV = \alpha I + \beta V \quad (28)$$

$$m_r = \text{mode}(r); m_g = \text{mode}(g); m_b = \text{mode}(b) \quad (29)$$

$$m = \frac{m_r + m_g + m_b}{3} \quad (30)$$

The mean value of the modal (MOM) values could then be used to enhance all three channels and preserve the colour relationships of all three channels or at worst, enhance the colour relationship rather than degrade it. However certain images yield zero values for these modal parameters for either red, green or blue channels. This skews the mean value once more leading to inconsistent results. In other words, performing a simple mean of modal values does not work well for all images, leading to spurious results when either $m_r = 0$, $m_g = 0$ or $m_b = 0$. Thus, we remove all zero entries and compute the mean of the remaining, non-zero entries. This ensures constant and consistent enhancement, while avoiding colour fading in all image enhancement results. Since the mode is more stable, the clustering of the colour component pixels (particles) around the mode should be consistent and this has been observed in other work [41,42].

5. Experiments and comparisons

The results in this section are presented to verify the claims made concerning the proposed colour enhancement/restoration algorithm (PA). The first part introduces the relevant metrics used in assessing image quality and how they are interpreted in results. The second part deals with the augmentation of the colour enhancement abilities of existing greyscale image-based contrast enhancement algorithms with PA. The third and fourth section compare PA with other relevant colour spaces and colour enhancement algorithms from the literature respectively. The fifth section discusses the justification for the substitution of the AMF within the EPF component of the SLIP-GUM algorithm for reduced computational time. The sixth section deals with attempts to adaptively determine the type and degree of degradation (unbalanced colour or uneven illumination) of the input image prior to processing.

5.1. Image quality metrics

The visual and quantitative results are presented to verify the claims made about the proposed algorithm. The quantitative measures for evaluation include the Colourfulness (C) [40], Colour Enhancement Measurement (EMEC) [43], Average Gradient (AG) and Hue Deviation Index (HDI) [44] and entropy. The relevant mathematical expressions for these measures are shown in Eqs. (31)–(36). These are in addition to computation of mean (μ) and standard deviation (σ) of the image, I in addition to the Contrast Enhancement Factor (CEF).

$$CEF = \frac{\sigma^2}{\mu}; \mu = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N I(i, j); \sigma^2 = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N [I(i, j) - \mu]^2 \quad (31)$$

$$\text{entropy} = \sum_i^n (-p_i * \log(p_i)) \quad (32)$$

$$AG = \frac{1}{MN} \sum_{x=1}^M \sum_{y=1}^N \sqrt{\frac{[I(x+1, y) - I(x, y)]^2 + [I(x, y+1) - I(x, y)]^2}{2}} \quad (33)$$

$$C = \sqrt{\sigma_\alpha^2 + \sigma_\beta^2 + 0.3 \sqrt{\mu_\alpha^2 + \mu_\beta^2}} \quad (34)$$

$$EMEC = \frac{1}{k_1 k_2} \sum_{k=1}^{k_1} \sum_{l=1}^{k_2} 20 \log \left[\frac{\max_{k,l}(I_R, I_G, I_B)}{\min_{k,l}(I_R, I_G, I_B)} \right] \quad (35)$$

$$HDI = \left[\frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N \min(|X(i, j) - Y(i, j)|, 1 - |X(i, j) - Y(i, j)|) \right] \times 100\% \quad (36)$$

For all relevant Eqs. in (31)–(36), M and N are the image dimensions depicting rows and columns respectively. For Eq. (31) the term σ^2 , is the variance and its square root is the standard deviation as mentioned earlier.

In Eq. (32), p_i is the normalized histogram of the image of bin element, i and n is the total number of bins and higher entropy implies that details are enhanced. For Eq. (33), I , is the processed image and any increase in AG implies increased contrast. In Eq. (34), $\alpha = R - G$, $\beta = \left(R + \frac{G}{2}\right) - B$, μ_α , σ_α , μ_β and σ_β are the mean and standard deviations of α and β respectively while R , G and B are the red, green and blue channels of either original or processed colour image and increase in colourfulness implies colour enhancement. Human-based subjective rankings can be objectively estimated using some defined metrics [40]. Thus, this metric is useful since it had a correlation of over 90% with human subjective evaluation results obtained from psychophysical experiments according to its originators [40]. Thus, we expect that increased colourfulness value of processed

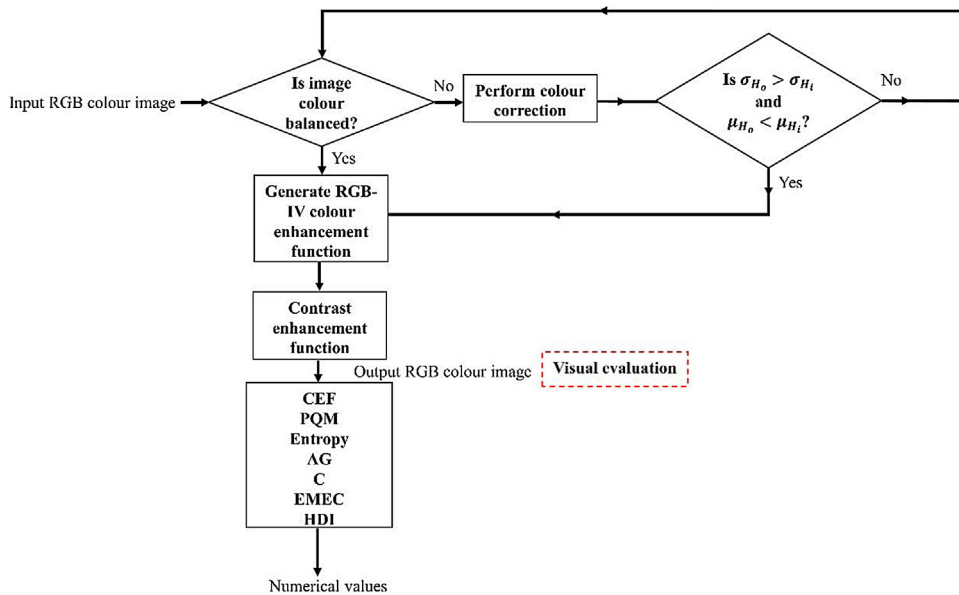


Fig. 7. Flowchart of proposed approach.

images would be seen as more colourful by most humans. Furthermore, results by Mukherjee show that colourfulness metric is not considerably affected by compression level though blockiness increases with increasing compression level [32]. This makes the colourfulness metric a robust and reliable parameter for assessment. In (35), the RGB colour image, $I = (I_R, I_G, I_B)$, while k_1 and k_2 are the size of the blocks chosen for subdivision of the image, k and l are the row and column indices of the image; also increased EMEC signifies improved colour. In Eq. (36), $Y(i, j)$ is the processed hue channel and $X(i, j)$ is the original hue channel of HSI or HSV transformed RGB colour image.

For the HDI, a lower value means that there is reduced colour distortion or deviation from the original colour image signifying good hue preservation properties (good cohesion among particles) and measures appropriate colourfulness value based on regulated dilation of particles. Additionally, the ratios of the values obtained from the processed and unprocessed images are obtained and provided for comparisons. If the ratios are less than 1, then there is degradation while ratio values greater than 1 imply improvement or enhancement. The ratio of the HDI is not computed since the metric utilizes the original and processed images in computation of the deviation parameter. The flowchart for the entire experimental system setup is shown in Fig. 7. The details of determination of colour balance of the input colour image and checking for adequate colour correction is explained in subsequent sections.

5.2. Colour enhancement of images processed with greyscale-based algorithms

This section presents results visually observed in images processed with the proposed colour enhancement scheme combined with existing illumination correction and contrast enhancement algorithms. We compare results using the *swan* image [5] processed with HSI, HSV, RGB and IV-based versions of the GHE, AHE, CLAHE, spatial HF (SHF) and frequency domain HF (FDHF), MSR, HS and GUM algorithms and shown in Fig. 8.

The visual and numerical results in Fig. 8 and Table 2 show that the proposed colour enhancement approach is consistent and improves colour rendition in the processed images. In Fig. 8, AHE, GHE and HS yield the most noticeable distortions in colour while the log-ratio (LR) and logarithmic image processing (LIP)-based approaches such as the Homomorphic filters (SHF & FDHF), MSR and LIP-GUM yield less distortion though more colour fading is observed. However, the HF and MSR algorithms yield the least distortion for the LIP-based approaches due to their excellent dynamic range compression abilities, while the MSR combines large scale and small scale features to preserve the multi-scale features of the image scene.

We also compare the proposed algorithm with the results of Ratio Rule, HSI version of the spatial Homomorphic filter, RGB-IV spatial HF, MSRCR and HSV-SLIP-GUM in Fig. 9 and results show that the algorithm yields better results than the other approaches, while being relatively less complex compared to the Ratio Rule and MSRCR versions. For example, the sky is much more colourful in Fig. 9(d) when compared with the other results while there is more contrast in the sky for SLIP-GUM result due to the CLAHE contrast enhancement component.

For the *swan* image used, the initial values for EMEC, CMA, mean (μ_i), standard deviation (σ_i), entropy (e_i) and average gradient (AG_i) are 13.43151, 15.63302, 52.18342, 62.46685, 6.564669 and 5.813808 respectively. The output values of the corresponding metrics are shown in Table 3. In both tables, the first row of each sub-table is for processing in RGB space without the proposed approach (RGB) while the second row in each sub-table is for processing in RGB space using the proposed algorithm (RGB-IV).

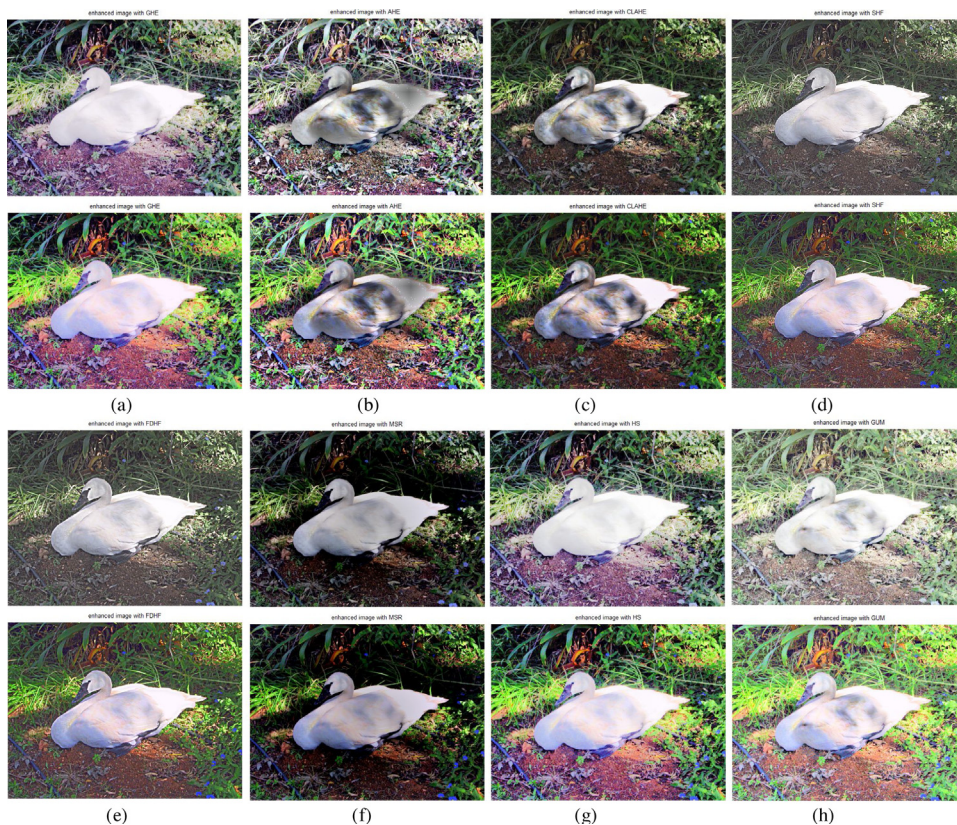


Fig. 8. Swan image processed with (a) GHE (b) AHE (c) CLAHE (d) SHF (e) FDHF (f) MSR (g) HS (h) GUM in RGB domain (first and third rows) and with proposed IV colour enhancement algorithm (second and fourth rows).

Table 2

Quantitative image quality measurement results for swan image processed with RGB and RGB-IV versions of the (a) GHE (b) AHE (c) CLAHE (d) SHF (e) FDHF (f) MSR (g) HS (h) GUM algorithms.

Version\image metrics	RC	F	EMEC	RM	RSD	RE	RAG	HDI	EMEC.2
RGB	2.345871	0.535439	2.073571	2.46279	1.148335	1.076786	3.067447	11.31385	27.85118
RGB-IV (a)	4.108481	0.597002	3.010375	2.320705	1.177058	1.21114	3.119015	5.534485	40.43387
RGB	2.465277	0.508135	3.831895	2.369004	1.097167	1.213991	4.309845	14.60396	51.46812
RGB-IV (b)	4.176792	0.546846	4.409189	2.215135	1.100608	1.206887	4.235839	7.046899	59.22205
RGB	1.628192	0.593445	1.225813	1.533186	0.953867	1.142119	2.278457	6.69303	16.46451
RGB-IV (c)	3.028428	0.654258	1.83028	1.45935	0.977134	1.140115	2.288706	4.032598	24.58342
RGB	1.36406	0.492213	0.946139	2.044878	1.003252	1.131727	2.924515	7.645274	12.70807
RGB-IV (d)	2.930466	0.553872	1.389627	1.933869	1.034947	1.144536	2.970556	3.637232	18.66479
RGB	1.306712	0.447208	0.701319	2.061137	0.960081	1.115542	2.326636	5.14373	9.419773
RGB-IV (e)	2.975797	0.512209	1.106271	1.943448	0.997723	1.137449	2.418773	2.448391	14.85889
RGB	1.648464	0.973854	–	1.126875	1.047574	0.960945	2.131063	9.087314	–
RGB-IV (f)	2.701197	1.000027	–	1.083761	1.041053	0.958091	2.066362	7.253675	–
RGB	2.433331	0.573	3.25958	2.44251	1.18303	1.218645	3.060409	10.83352	43.78107
RGB-IV (g)	4.219846	0.635281	3.756287	2.304472	1.209953	1.215316	3.103987	5.132268	50.45259
RGB	1.967059	0.249363	0.692045	3.140091	0.884885	1.169709	2.86375	7.819348	9.295204
RGB-IV (h)	4.464721	0.334464	1.06233	2.936177	0.990983	1.173846	3.102782	3.015748	14.2687

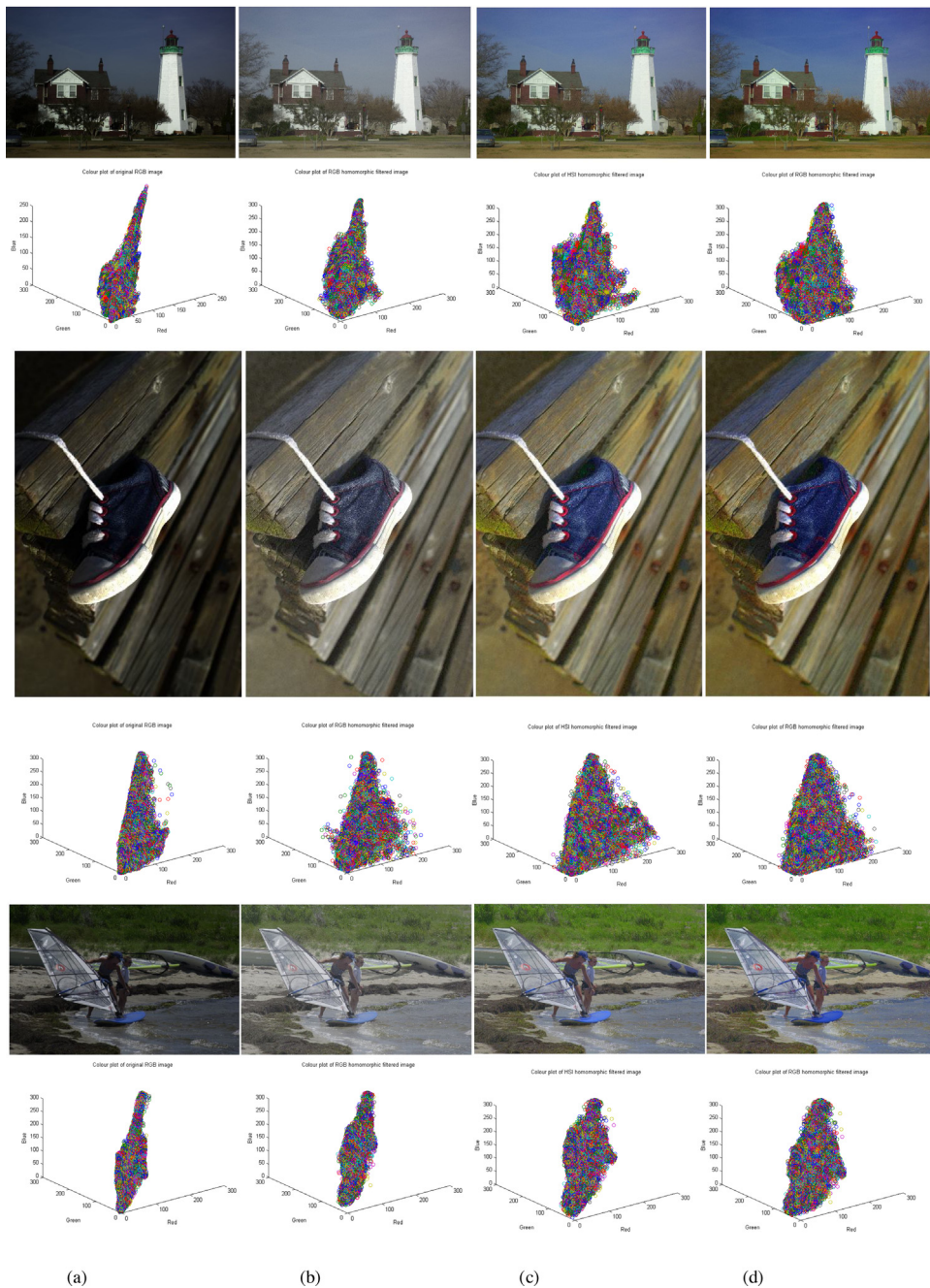


Fig. 9. (a) *Light-house, shoe and surfers* images processed with (b) RGB HF (c) HSI-HF and (d) RGB-IV-HF and respective colour plots.

The results in Table 2 show that for colour enhancement metrics such as relative colour enhancement (RC), relative CEF (F), relative EMEC (REMEC), the proposed colour enhancement algorithm gives consistently higher values than the standard RGB-based approaches. The colourfulness and EMEC values have increased while the HDI value has dropped considerably, indicating that the colour enhancement algorithm has reduced the hue deviation and avoided colour distortion effect of the various algorithms, thus the lower HDI can be seen as better cohesion of colour components.

In general, the proposed scheme yields improved colour rendition when combined with grey-scale based enhancement algorithms for processing RGB colour images.

Table 3

Quantitative output image quality measurement results for swan image processed with RGB and RGB-IV versions of the (a) GHE (b) AHE (c) CLAHE (d) SHF (e) FDHF (f) MSR (g) HS (h) GUM algorithms.

Version\image metrics	CMB	mu.o	sigma.o	ent.o	AG.o
RGB	36.67305	128.5168	71.73287	7.068746	17.83355
RGB-IV	64.22798	121.1023	73.52713	7.950732	18.13335
(a)					
RGB	38.53974	123.6227	68.53655	7.969447	25.05661
RGB-IV	65.2959	115.5933	68.75151	7.922814	24.62635
(b)					
RGB	25.45356	80.00689	59.58507	7.497631	13.24651
RGB-IV	47.34349	76.15386	61.03852	7.484477	13.3061
(c)					
RGB	21.32438	106.7087	62.67002	7.42941	17.00257
RGB-IV	45.81204	100.9159	64.64988	7.513499	17.27024
(d)					
RGB	20.42786	107.5572	59.97325	7.323166	13.52661
RGB-IV	46.5207	101.4158	62.32463	7.466973	14.06228
(e)					
RGB	25.77048	58.80421	65.43866	6.308283	12.38959
RGB-IV	42.22787	56.55438	65.03129	6.289551	12.01343
(f)					
RGB	38.04033	127.4585	73.90013	8	17.79263
RGB-IV	65.96896	120.2552	75.58197	7.978148	18.04598
(g)					
RGB	30.75108	163.8607	55.27601	7.67875	16.64929
RGB-IV	69.7971	153.2198	61.90356	7.705911	18.03898
(h)					

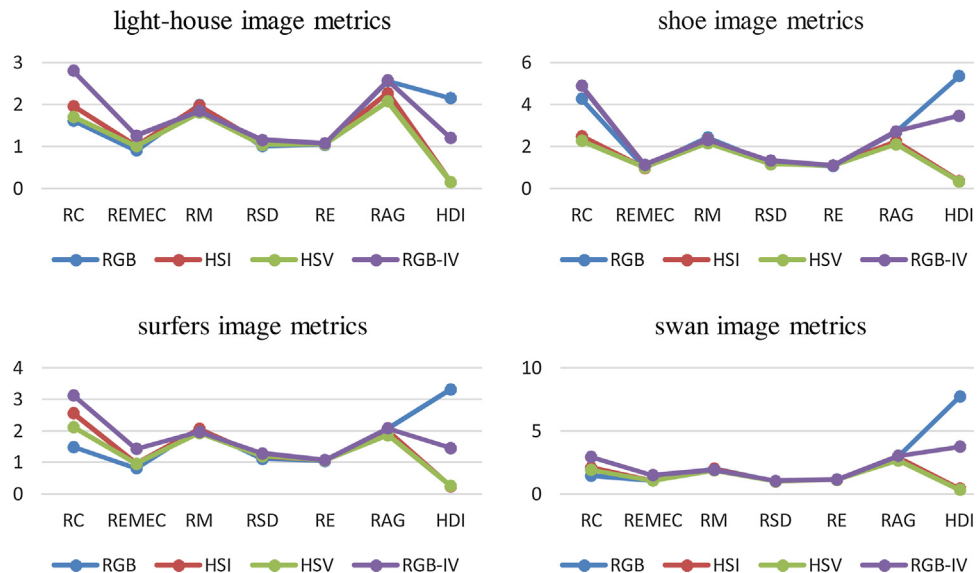


Fig. 10. Image metric plots for processed *Light-house*, *Shoe*, *surfers* and *swan* images using RGB-HF, HSI-HF, HSV-HF and RGB-IV-HF.

5.3. Comparisons with relevant colour spaces from the literature

The proposed approach is also compared with the relevant colour spaces in this section and results indicate improved colourfulness compared to nonlinear colour spaces such as HSI and HSV systems. Due to space constraints, we only present a sample of the results for the HF algorithm instance. The image results and corresponding RGB plots are shown in Fig. 9, while the plots of metrics of images processed with alternate colour space implementations of the HF are shown in Fig. 10. In some cases, (*shoe* image [5] in Fig. 9) the RGB-IV result is visually indistinguishable from the HSI-based scheme while in other cases, it yields clearly richer colours and saturation without bleeding (*light-house* and *surfers* images [5]). The colour



Fig. 11. (a) image from [7] (b) HF + ratio rule [6,7] (c) HSI-HF from [8] (d) HF+ RGB-IV (PA) (e) MSRCR (f) HSV-SLIP-GUM.

fading is clearly observed in the standard RGB result of the HF in Fig. 9 and which has the lowest RC/CEF and REMEC values in the plots in Fig. 10.

Also, observing the RGB colour plots in Fig. 9, there is less spreading of the channels in relation to each other using RGB compared to HSI and RGB-IV-based schemes. We illustrate this point with the plots of RC/CEF, HDI and EMEC in Fig. 10 for the processed images. The HDI can be used as a measure of the degree of spread since too much dilation leads to colour distortion, while excessive contraction results in fading corresponding to very high HDI (RGB result), while a moderate HDI indicates adequate dilation (RGB-IV result) and minimal HDI yields the ideal balance, as observed in results using HSI and HSV colour spaces.

5.4. Comparisons with other algorithms from the literature

We also compare the proposed scheme with other colour enhancement algorithms from the literature such as MSRCR, Ratio rule-based HF, HSI-HF, HSV-GUM and DCT-based CES versions. The results in Fig. 11 indicate that best results are obtained with PA compared to the DCT-based method [32] and RGB, HSI variants of the HF for the *surfer*, *shoe* and *light house* images from [5].

Additionally, we also observe that PA gives the highest colourfulness of all the processed image results in Fig. 11 and Table 4 compared to other approaches from the literature, based on CEF/RC values. It should be noted that several of these other approaches still require processing in the HSV/HSI/YUV/YCbCr colour space to improve colour rendition results. Conversely, PA can replicate and improve on these results since it can mimic any of the colour spaces without performing the full forward or reverse transformations.

The expression for estimating computational complexity given by [32] as; $aA + mM + eE$ is employed, where a is the number of additions, m is the number of multiplications and e is the number of exponential operations per pixel. However, the estimates from [32] do not include divisions and logarithms or incorporate the iterative nature of the DCT-based CES algorithm. We include divisions and logarithmic operations per pixel as d and l in the amended expression given as; $aA + mM + eE + dD + lL$ omitting max/min, modal, mean, trigonometric and compare operations. We compare structural complexity of algorithms in Table 5. Based on results, the proposed scheme contributes minimal multiplication operations, though it has a relatively large number of division operations per pixel. Only the HSI and HSV forward conversion opera-

Table 4

Comparison between CES variants with values from [32] amended with computed values for the HF variants and the proposed approach combined with HF.

Algorithm	CEF/RC		
	surfers	light-house	shoe
TW-CES	1.74	1.71	1.70
DRC—CES	1.39	1.27	1.44
SF-CES	1.45	1.40	1.44
TW-CES-BLK	1.74	1.71	1.71
DRC—CES-BLK	1.39	1.28	1.44
SF-CES-BLK	1.45	1.40	1.44
RGB-HF	1.35	1.01	1.23
HSI-HF	2.35	1.84	2.11
RGB-IV-HF (PA+HF)	2.64	2.26	2.54

Table 5

Comparison of computational complexity and run-time between CES variants with values from [32] amended with computed values for the relevant colour spaces and the proposed approaches.

Algorithm	Computational complexity
AR	1E+1M [32]
MCE	2.19M+1.97A [32]
MCEDRC	0.03E+3.97M+2A [32]
TW-CES	0.02E+4.02M+1.05A [32]
DRC—CES	0.05E+4M+1.08A [32]
SF-CES	0.03E+4.02M+1.06A [32]
HSI	1E+9A+3D
HSV	5A+4D
MSRCR	18E+1866378M+8156703A [32]
RGB-IV	1E+1M+3A+3D

tions are taken into account in the calculation of their computational complexity. It is also not clear whether the algorithms proposed by [32] included the calculations of the operations for the colour space conversions utilised in their approach. It is also observed that the MSRCR has the highest computational complexity as noted in [32].

In summary, PA attempts to address colour distortion and improve consistency of results, while introducing automation with moderate complexity. Having made the case for the PA, we proceed with reducing the run-time of the earlier mentioned GUM algorithm.

5.5. Edge preserving filter (EPF) replacement in GUM

The AMF, which functions as the EPF component in the GUM algorithm is of considerable computational complexity and this adds to the increased run-time of the GUM algorithm. Thus, for the EPF component, we explore three contemporary filters that surpass the edge preserving properties of the AMF based on PSNR, SSIM, MAE and MSE measurements [45] (see also [46]). These three filters include;

- The Noise Adaptive Fuzzy Switching Median Filter (NAFSMF) [47]
- Entropy guided Switching Trimmed Mean Deviation-boosted Anisotropic Diffusion filter (STMDF-AD) [45]
- The Efficient Weighted Average (EWA) Filter [48]

The results are shown in Fig. 12 and the *mesa* image from [11] was processed using the modified GUM algorithm with the aforementioned filters as EPF replacements. For the NAFSMF, the processing of the canyon image is about 0.037363 s, while the processing time for AMF is about 6.561111 s and for the STMDF-AD, the processing time is about 4.194520 s for one iteration. Lastly, for the EWA filter, the processing time is about 0.336881 seconds. Thus based on results, the NAFSMF is the least computationally complex filter and also the fastest while yielding visually similar results to the AMF and EWA filter (which is the second fastest edge preserving filter). The STMDF-AD filter is the second slowest and most complex of the chosen EPFs. Also, its results do not work well in some regions of the processed image since it performs a thresholding effect though there is evidence of increased colour enhancement (Fig. 13(a)). Either the EWA filter or the NAFSMF can be selected as a EPF replacement, drastically reducing the run-time of the LIP-GUM and SLIP GUM algorithms as a result. We also replace the HSV component with the PA in the GUM algorithm and compare performance.

A summary of results is presented in Table 6 for processing the RGB colour *swan* image with dimensions 385×513 . Results show that the STMDF-AD yields the highest values for F and RSD while the NAFSMF yields the best EMEC, RM, RAG values and lowest runtimes. The RGB-IV-based GUM yields the highest RC values and lower runtimes than the HSV-based GUM algorithm. However, the HDI is higher than for the HSV, since hue is not considerably modified in the HSV colour space. Nevertheless, the HDI varies depending on the image being processed, while the RGB-IV scheme yields better results than the



Fig. 12. (a) STMDf-AD ($dt = 0.25$, 1 iteration) (b) NAFSMF (c) EWA (d) AMF.



Fig. 13. (a) RGB-IV and (b) HSV versions of the SLIP-GUM algorithm using the NAFSMF as the edge preserving filter (EPF).

standard RGB version and is much faster than the HSI and HSV versions of the SLIP-GUM algorithm (due to parallelization). Moreover, the image results from both HSV and RGB-IV versions of the SLIP-GUM in Fig. 13 are quite similar and bolded values in Table 6 indicate better results.

With the RGB-IV and the EPF substitutions, the run-time of the SLIP-GUM algorithm is drastically reduced, while colour and contrast enhancement results are more or less maintained. Thus, the proposed colour enhancement algorithm allows the option of processing in the RGB space, while emulating the visual, colourful results of the HSI/HSV colour spaces and considerably reducing execution time of the algorithms it augments.

5.6. Classification of image colour quality attributes and limitations of PA

It is important to note that although the proposed colour enhancement works well for balanced colour images, it is not suitable for images with haze or dominant colour casts such as underwater images. These images are predominantly blue or green since the blue light has the shortest wavelength and penetrates at greater depths than the other colours [49]. Based on this fact, we attempt to classify images based on the perceptual metrics and also on the modal ratio values of the various images obtained in Table 1. The aim is to understand the limitations of the proposed approach and to predict which images suffer from colour imbalance in order to automatically correct them prior to processing especially those suffering

Table 6

Quantitative image quality measurement and runtime results for swan image processed with (a) HSV and (b) RGB-IV versions of the SLIP-GUM algorithm using various edge preserving filters (EPF).

Metrics\EPF	AMF	NAFSMF	STMDF-AD	EWA
RC	2.7594	2.799	2.402	2.799
F	0.6432	0.6298	0.7266	0.6299
EMEC_score	3.8969	3.9786	3.9218	3.9786
RM	1.3908	1.4228	1.2712	1.4228
RSD	0.9458	0.9466	0.961	0.9467
RE	1.1321	1.1379	1.1097	1.1379
RAG	1.9031	2.1289	1.7817	2.1289
HDI	0.2669	0.2607	0.2914	0.2608
EMEC_1	13.4315	13.4315	13.4315	13.4315
EMEC_2	52.3409	53.4384	52.6755	53.4384
sim time (s)	5.286669	0.158715	3.520937	0.533247
(a)				
RC	3.121567	3.133499	2.732197	3.133552
F	0.647371	0.631099	0.721767	0.631226
EMEC_score	1.357936	1.689079	1.598473	1.650713
RM	1.491472	1.532792	1.355827	1.532674
RSD	0.982617	0.983536	0.989238	0.983597
RE	1.141315	1.149279	1.119862	1.149288
RAG	1.997641	2.311309	1.92015	2.311256
HDI	4.945912	4.102212	3.929718	4.093378
EMEC_1	13.4315	13.4315	13.4315	13.4315
EMEC_2	18.23913	22.68688	21.4699	22.17156
sim time (s)	5.21759	0.024393	3.245898	0.184838
(b)				

from uneven illumination. It should be noted that though several illumination correction algorithms can perform colour correction on some images, their results are usually not consistent. Also, some images which do not visually exhibit this colour imbalance may still be affected by colour correction algorithms, especially if their dynamic range is not fully utilized. Thus a colour correction stage should be applied only when necessary and we perform some experiments to enable us obtain more information about these types of images. We utilize the mean and the standard deviation of the hue, saturation and value components to help quantify the amount of clustering and dispersion around the mean value for each of the components. It may be deduced that such high variation is an indication of greatly increased contrast as observed in previous works [4,32]. Based on the preliminary experiments we were able to infer three classes of colour images namely;

- Dark colour images with uneven illumination (and dominant colour cast)
- Colour images with shadows/uneven illumination
- Faded colour images with dominant colour cast

Further extensive experiments were performed involving 195 natural images of various types of illumination used by various authors in the literature (see image references). Based on some prior simplistic assumptions; images with the histograms skewed to the right of the distribution are likely to be faded or bright images while those skewed to the left are likely to be the dark images. For the clustered dataset in Fig. 14, lower mean and standard deviation values imply dark images while higher mean and standard deviation values imply brighter images. However, this scheme appears too ambiguous and may make it difficult to properly classify some images. Thus, the images are converted to greyscale from which a binary mask is generated, then the area of light vs black pixels is computed to determine if the image is dark or light.

Based on the plot in 14(c), it is visually clear that the proportion of black pixels is always greater than the proportion of white pixels, thus confirming that all the tested images can be visually perceived as predominantly dark. However, there are varying degrees of proportions based on the different peaks of both white and dark pixel counts, thus the second class of images where the area of black and white pixels is roughly equal. This makes it difficult to accurately predict since some images may have more dark pixels than bright ones but will have them spread out evenly, thus having the appearance of a reasonably bright image with shadow regions. This could lead to the image being misclassified as a dark image and processed accordingly.

Other factors include the thresholding function used and accuracy of the selected threshold value, which may move additional pixels below or above the selected threshold. The particular scheme utilized for this experiment is Otsu's algorithm [50], which computes an adaptive threshold. Experiments were repeated for images with dominant colour cast to determine these features numerically, and the results are presented in Fig. 15. For the plot in Fig. 15(a), the 195 images have their hue components measured for mean and standard deviation and show high values for both quantities with some correlation. However, in the case of 35 underwater images with dominant blue colour cast, the mean and standard deviations of the hue components are quite low and appear to be uncorrelated in 15(b). The images are then colour corrected and have their respective hues measured once more using the mean and standard deviations in 15(c). There is an observed general increase

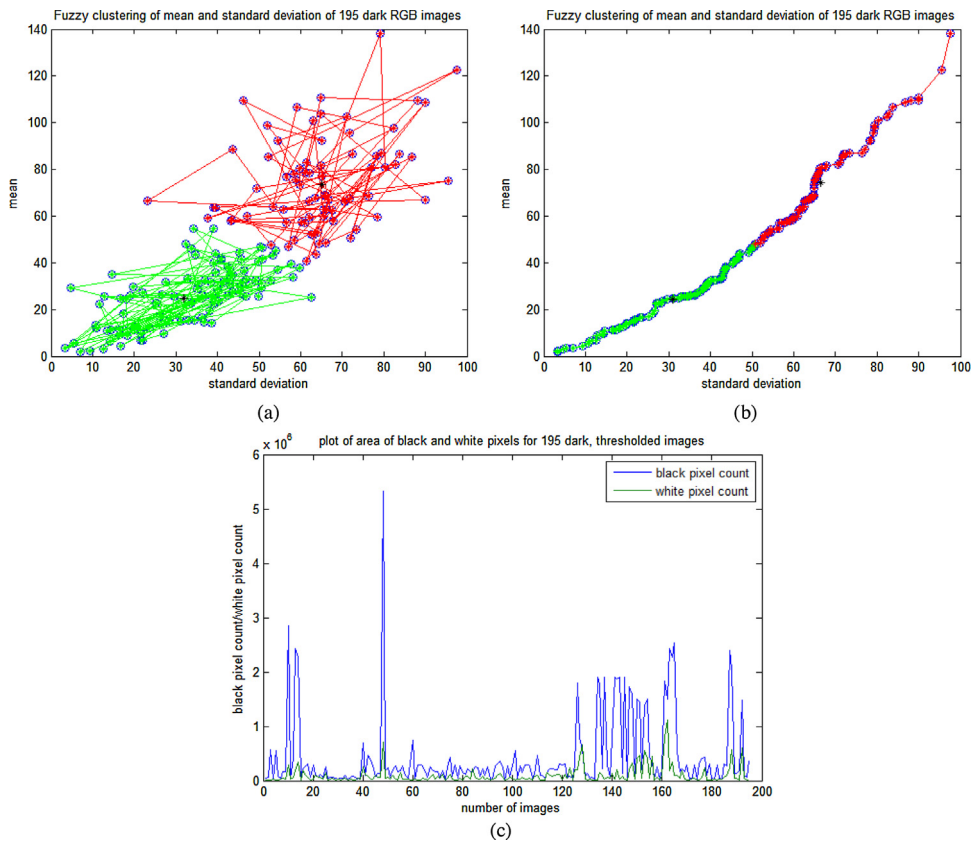


Fig. 14. Fuzzy clustering of 195 dark colour image mean brightness and standard deviation using (a) unordered and (b) ordered feature dataset (c) plot of black and white pixel count of 195 dark, thresholded images.

in standard deviation, and variance (contrast) while the mean generally has decreased. The proposed colour enhancement scheme is applied to dark images with dominant colour casts by comparing the mean and standard deviation values of the hue components. An increase or decrease in standard deviation will indicate the effectiveness of the colour enhancement for such images. However, the scheme mostly yields over-saturation of colour corrected images since these do not have faded colours. Thus, it is generally ineffective for underwater images, which are better served by previously developed algorithms in the literature.

The findings indicate that a mainly colour correction or white balancing algorithm does not have considerable contrast enhancement or illumination correction effect on images, which are not suffering from dominant colour cast. However, utilizing a strong contrast enhancement algorithm (which depends on the input image statistics such as standard deviation or mean) will drastically alter the results, considerably modifying the image histogram and consequently, the pixel intensity values in each of the R, G and B channels [3,51,52].

The HSI and HSV-based algorithms cannot perform proper colour correction since their inherent strengths of hue preservation and decoupling, works against colour correction, which must be performed in RGB space for images with colour cast. However, for contrast enhancement and illumination correction, only the intensity component (for images without colour cast problems) is usually processed.

In summary, an ideal colour correction algorithm should only affect the hue and (or) saturation component and not distort the RGB colour relationships of the three channels. Conversely, an ideal contrast enhancement and illumination correction algorithm should modify only the pixel intensities and eliminate uneven illumination without distortion of the hue/saturation components. However, this is rarely achievable in practice due to the non-linear nature of most contemporary algorithms, thus the need for processing in the HSI and HSV colour spaces. It has been shown that the proposed scheme helps to somewhat mitigate this imperfection of contrast enhancement and illumination correction algorithms enabling improved colour processing in RGB space. However, for images with dominant colour cast, the proposed algorithm is unsuitable due to initial incorrect colour composition of such images. Thus, by extracting features that help describe this dominance in such images, a colour dominance degree detection scheme can be used to determine suitability for such images.

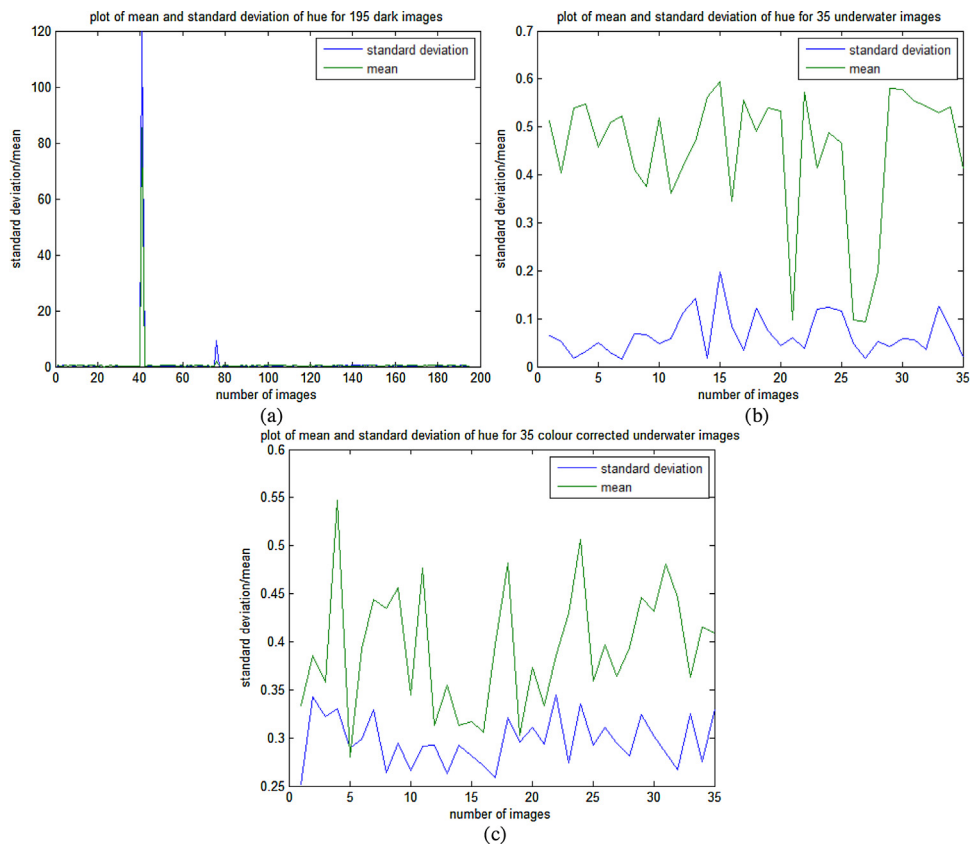


Fig. 15. Plot of mean and standard deviation for hue component of (a) 195 dark colour images (b) 35 underwater colour images with dominant colour cast and (c) corresponding colour corrected outputs using GOC2 algorithm.

6. Discussions and possible application areas

Based on experiments and observable results, the proposed approach eliminates colour fading and distortions usually associated with application of greyscale image enhancement algorithms to RGB colour images. The proposed approach is fast and naturally parallelizable while being relatively easier to implement in FPGA hardware using either fixed or floating point scheme, yielding visually comparable results similar to HSI and HSV-based approaches. The colour bleeding effect observed with processing certain images in the HSI colour space has been mitigated by the combined I–V and adaptive approach.

Future possibilities include optimized implementation of the colour enhancement system in hardware for comparison with previous work utilizing HSI and HSV colour spaces in terms of both speed, hardware resource usage and performance. Ultimately, the proposed colour enhancement algorithm due to its generality, can be applied in digital colour photography, tone mapping techniques and remote sensing. Future work will explore the application of the algorithm to other areas in more detail.

7. Conclusion

A fast and generalized colour enhancement model has been proposed in this work and shows performance gains in image colour fidelity and reduced complexity. The proposed approach can be modified to mimic the results of any colour space without performing the full transformation in that colour space. This results in cost savings in hardware and faster operation on devices with limited memory and real-time constraints. The colour enhancement algorithm is non-iterative and based on the colourfulness and HDI measure yields excellent and consistent colour enhancement compared to naïve RGB implementations. An adaptive pre-colour correction stage is performed for (dark) images suffering from dominant colour cast prior to colour enhancement and illumination correction as determined from the modal ratio values and statistical parameters of the hue component.

Furthermore, we have also improved the effectiveness of the SLIP-GUM algorithm by substitution of the colour and EPF component drastically reducing execution time while maintaining similar results obtained in HSV/HSI domain using the proposed colour enhancement scheme.

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